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Contents

Volume 4, Number 6, November/December 2024

Editorial: Barriers to Extracting and Harmonizing Glaucoma Testing 100621 **Data: Gaps, Shortcomings, and the Pursuit of FAIRness**

Niloofar Radgoudarzi, BS, Shahin Hallaj, MD, Michael V. Boland, MD, PhD, Brian Stagg, MD, Sophia Y. Wang, MD, MS, Benjamin Xu, MD, PhD, Swarup S. Swaminathan, MD, Eric N. Brown, MD, PhD, Aiyin Chen, MD, Catherine Q. Sun, MD, Dilru C. Amarasekera, MD, Jonathan S. Myers, MD, Murtaza Saifee, MD, William Halfpenny, MB BCHir, MEng, Keri Dirkes, MPH, Linda Zangwill, PhD, Kerry E. Goetz, PhD, MS, Michelle Hribar, PhD, MS, Sally L. Baxter, MD, MSc

Original Articles

Impact of Tumor Pigmentation in 6934 Patients with Uveal Melanoma 100585 **at a Single Center**

Samuel J. Goldstein, BSPH, Ferris Bayasi, MD, George Thomas, MD, Matthew Barke, MD, Michael K. Nguyen, BA, Samantha Pastore, BS, Carol L. Shields, MD

This analysis of 6934 patients with uveal melanoma (UM) found that pigmented UMs demonstrated greater rates of metastasis (Kaplan–Meier analysis at 2, 5, and 10 years) compared with partially pigmented and nonpigmented UMs.

(Continued)

On the cover: Cover image by Ralph C. Eagle, Jr., MD (Department of Pathology, Wills Eye Hospital, Philadelphia, PA) was the First Place winner in the Photo/Electron Micrography category at the Ophthalmic Photographers' Society 2023 Scientific Exhibition (www.opsweb.org).



2666-9145(202411/12)4:6;1-H

Contents

(Continued)



Retinal Microstructural Changes Reflecting Treatment-Associated Cognitive Dysfunction in Patients with Lower-Grade Gliomas 100577

Arina Nisanova, BA, Ashutosh Parajuli, BS, Bhavna Antony, PhD, Orwa Aboud, MD, PhD, Jinger Sun, MD, PhD, Megan E. Daly, MD, Ruben C. Fragoso, MD, PhD, Glenn Yiu, MD, PhD, Yin Allison Liu, MD, PhD

This study of 16 patients with lower-grade gliomas shows the potential application of retinal microstructural changes as proxy of treatment-associated cognitive dysfunction (TACD). Tumor pathology and location may further increase susceptibility to TACD.



An in vitro 3-dimensional Collagen-based Corneal Construct with Innervation Using Human Corneal Cell Lines 100544

Mohammad Mirazul Islam, PhD, Amrita Saha, PhD, Farzana Afrose Trisha, BS, Miguel Gonzalez-Andrades, MD, PhD, Hirak K. Patra, PhD, May Griffith, PhD, MBA, James Chodosh, MD, MPH, Jaya Rajaiya, PhD

Corneal research largely depends on animal models. To reduce animal use and improve translation of *in vitro* data to the clinic, the authors developed a 3-dimensional corneal model which closely mimics the human cornea.

Prevalence and Associated Factors of Cataract, Cataract Surgery and Postoperative Outcome in an Old Population in Russia: The Ural Very Old Study 100545

Mukhamram M. Bikbov, MD, PhD, Gyulli M. Kazakbaeva, MD, Songhomitra Panda-Jonas, MD, Ellina M. Lakupova, MD, Albina A. Fakhretdinova, MD, Azaliia M. Tuliakova, MD, Jost B. Jonas, MD

In a multiethnic study cohort aged >85 years in Russia, prevalence of cataract-related moderate-to-severe vision impairment (28.2%) and blindness (2.9%) were high despite a high cataract surgery rate (52.5%).



Safety Results for Geographic Atrophy Associated with Age-Related Macular Degeneration Using Subretinal Cord Blood Platelet-Rich Plasma 100476

Stanislao Rizzo, MD, Maria Cristina Savastano, MD, PhD, Benedetto Falsini, MD, Patrizio Bernardinelli, MD, Francesco Boselli, MD, Umberto De Vico, MD, Matteo Mario Carlà, MD, Federico Giannuzzi, MD, Claudia Fossataro, MD, Gloria Gambini, MD, Emanuele Crincoli, MD, Silvia Ferrara, MD, Matteo Ripa, MD, Raphael Killian, MD, Clara Rizzo, MD, Caterina Giovanna Valentini, MD, PhD, Nicoletta Orlando, MSc, PhD, Giorgio Placidi, CO, Luciana Teofili, MD, PhD, Alfonso Savastano, MD, PhD

This study reports the results of a case-control study regarding the subretinal injection of cord blood platelet-rich plasma in progressive macular geographic atrophy. Data showed safety of cord blood platelet-rich plasma at 12-month follow-up, but its efficacy cannot be determined.

(Continued)

Contents

(Continued)



Phase I Study of Tivozanib Eye Drops in Healthy Volunteers and Patients with Neovascular Age-Related Macular Degeneration 100553

Fumi Gomi, MD, PhD, Tomohiro Iida, MD, PhD, Ryusaburo Mori, MD, PhD, Shinya Horita, PhD, MBA, Hiroaki Nakamura, MS, Yu Nakajima, MMath, Ayako Shiokawa, BPharm, Kanji Takahashi, MD, PhD

This phase I study evaluated the safety and pharmacokinetics of tivozanib eye drops in healthy volunteers and neovascular age-related macular degeneration patients. No grade ≥ 3 adverse events occurred, and the safety profile was favorable.

Reticular Pseudodrusen: Impact of Their Presence and Extent on Local Rod Function in Age-Related Macular Degeneration 100551

Himeesh Kumar, MBBS(Hons), PhD, Robyn H. Guymer, MBBS, PhD, Lauren A.B. Hodgson, MPH, Xavier Hadoux, MEng, PhD, Maxime Jannaud, MEng, Peter van Wijngaarden, MBBS(Hons), PhD, Chi D. Luu, PhD, Zhichao Wu, BAppSc(Optom), PhD

This study observed that local rod-mediated visual function was associated with the overall, rather than local, extent of reticular pseudodrusen present in eyes with large drusen in a cohort with intermediate age-related macular degeneration.



Quantifying Changes on OCT in Eyes Receiving Treatment for Neovascular Age-Related Macular Degeneration 100570

Gabriella Moraes, MD, MSc, Robbert Struyven, MD, Siegfried K. Wagner, BMBCh, FRCOphth, Timing Liu, BA, David Chong, MBBChir, Abdallah Abbas, iBSc, MBBS, Reena Chopra, BSc, Praveen J. Patel, MD, FRCOphth, Konstantinos Balaskas, MD, Tiarnan D.L. Keenan, BM BCh, PhD, Pearse A. Keane, MD, FRCOphth

The authors report findings from a deep learning algorithm that automatically quantifies multiple OCT features at multiple time points in patients treated for neovascular age-related macular degeneration. Raw data are openly available for future research.



Artificial Intelligence-Based Disease Activity Monitoring to Personalized Neovascular Age-Related Macular Degeneration Treatment: A Feasibility Study 100565

Zufar Mulyukov, PhD, Pearse A. Keane, FRCOphth, MD, Jayashree Sahni, FRCOphth, MD, Sandra Liakopoulos, MD, Katja Hatz, MD, Daniel Shu Wei Ting, MD, PhD, Roberto Gallego-Pinazo, MD, PhD, Tariq Aslam, PhD, DM(Oxon), Chui Ming Gemmy Cheung, FRCOphth, MD, Gabriella De Salvo, FRCOphth, MD, Oudy Semoun, MD, Gábor Márk Somfai, MD, PhD, Andreas Stahl, MD, Brandon J. Lujan, MD, Daniel Lorand, MSc

A digital analysis tool based on artificial intelligence has been developed and evaluated against the majority vote from a panel of retina specialists to assess disease activity in patients with neovascular age-related macular degeneration.

(Continued)

Contents

(Continued)



Artificial Intelligence to Facilitate Clinical Trial Recruitment in Age-Related Macular Degeneration 100566

Dominic J. Williamson, MSc, Robbert R. Struyven, MD, Fares Antaki, MD, Mark A. Chia, MD, Siegfried K. Wagner, MD, PhD, Mahima Jhingan, MBBS, Zhichao Wu, PhD, Robyn Guymier, MBBS, PhD, Simon S. Skene, PhD, Naaman Tammuz, PhD, Blaise Thomson, PhD, Reena Chopra, PhD, Pearse A. Keane, MD

This study illustrates the potential of artificial intelligence in enhancing clinical trial recruitment for geographic atrophy through deep learning analysis of OCT scans, identifying eligible patients according to trial-specific imaging criteria.

Deep Learning–Based Clustering of OCT Images for Biomarker Discovery in Age-Related Macular Degeneration (PINNACLE Study Report 4) 100543

Robbie Holland, MEng, Rebecca Kaye, MD, Ahmed M. Hagag, MD, Oliver Leingang, PhD, Thomas R.P. Taylor, MD, Hrvoje Bogunović, PhD, Ursula Schmidt-Erfurth, MD, Hendrik P.N. Scholl, MD, Daniel Rueckert, PhD, Andrew J. Lotery, MD, Sobha Sivaprasad, MD, Martin J. Menten, PhD

The authors developed a self-supervised method to automatically identify novel biomarkers for age-related macular degeneration in large datasets of unlabeled OCT images. The authors find that discovery-based artificial intelligence systems could accelerate clinical research into novel prognostic biomarkers.



Improving the Identification of Diabetic Retinopathy and Related Conditions in the Electronic Health Record Using Natural Language Processing Methods 100578

Keith Harrigan, MS, Diep Tran, MSc, Tina Tang, MD, Anthony Gonzales, OD, Paul Nagy, PhD, Hadi Kharrazi, MD, PhD, Mark Dredze, PhD, Cindy X. Cai, MD, MS

The authors developed a natural language processing framework that identifies diabetic retinopathy and related conditions in electronic health records with better precision and recall than diagnostic codes alone.



Demographic and Metabolic Risk Factors Associated with Development of Diabetic Macular Edema among Persons with Diabetes Mellitus 100557

Rachana Haliyur, MD, PhD, Shikha Marwah, MS, Shreya Mittal, MS, Joshua D. Stein, MD, Anjali R. Shah, MD, on behalf of the SOURCE Consortium

In nearly 50 000 individuals with diabetes, Black race and modifiable risk factors of blood pressure and glycosylated hemoglobin predicted new diabetic macular edema. Other systemic markers of dysmetabolism such as obesity and dyslipidemia did not.

(Continued)

Contents

(Continued)



Comparison of Diagnosis Codes to Clinical Notes in Classifying Patients with Diabetic Retinopathy 100564

Sean Yonamine, MPH, Chu Jian Ma, MD, PhD, Rolake O. Alabi, MD, PhD, Georgia Kaidonis, MBBS, PhD, Lawrence Chan, MD, Durga Borkar, MD, Joshua D. Stein, MD, MS, Benjamin F. Arnold, PhD, Catherine Q. Sun, MD

The authors developed a rule-based natural language processing algorithm to identify diabetic retinopathy (DR) severity using clinical notes. The notes algorithm performed better than diagnosis codes for proliferative DR classification.



The Effect of Glucagon-Like Peptide-1 Receptor Agonists on Diabetic Retinopathy at a Tertiary Care Center 100547

Julia H. Joo, MPH, Neha Sharma, MPH, Jacqueline Shaia, BS, Anna K. Wu, MD, Mario Skugor, MD, Rishi P. Singh, MD, Aleksandra V. Rachitskaya, MD

GLP-1 receptor agonists (GLP-1RA) are implicated in diabetic retinopathy development and progression. After propensity score matching in 981 patients, the authors did not identify greater worsening on GLP-1RAs compared with the control, SGLT-2 inhibitors.



Liquid Biopsy for Proliferative Diabetic Retinopathy: Single-Cell Transcriptomics of Human Vitreous Reveals Inflammatory T-Cell Signature 100539

Rachana Haliyur, MD, PhD, David H. Parkinson, MS, Feiyang Ma, PhD, Jing Xu, PhD, Qiang Li, MD, PhD, Yuanhao Huang, Lam C. Tsoi, PhD, Rachael Bogle, MS, Jie Liu, PhD, Johann E. Gudjonsson, MD, Rajesh C. Rao, MD

Single-cell analysis of vitreous cells collected during vitrectomy for proliferative diabetic retinopathy revealed that activated T cells make up the primary vitreous cell-type; such cells comprise a far lesser proportion in the peripheral blood.

Clinical Profile and Ocular Morbidities in Patients with Both Diabetic Retinopathy and Uveitis 100511

Ayushi Mohapatra, DNB, Sridharan Sudharshan, MS, Parthoprattim Dutta Majumder, MS, Janani Sreenivasan, DNB, Rajiv Raman, MS, DNB

A higher prevalence of late stages of diabetic retinopathy, uveitic, and neovascular complications and a poorer visual outcome was noted in eyes with coexisting uveitis and diabetic retinopathy.

(Continued)

Contents

(Continued)



Comparative Analysis of Vision Transformers and Conventional Convolutional Neural Networks in Detecting Referable Diabetic Retinopathy 100552

Jocelyn Hui Lin Goh, BEng, Elroy Ang, BEng, Sahana Srinivasan, BEng, Xiaofeng Lei, MSc, Johnathan Loh, MEng, Ten Cheer Quek, BEng, Cancan Xue, PhD, Xinxing Xu, PhD, Yong Liu, PhD, Ching-Yu Cheng, PhD, Jagath C. Rajapakse, PhD, Yih-Chung Tham, PhD

Exploring vision transformers and convolutional neural networks, this study compares their performance in detecting referable diabetic retinopathy using fundus photographs. Findings can provide valuable insights for optimizing retinal disease detection methodologies.



Factors Associated with Missing Sociodemographic Data in the IRIS[®] (Intelligent Research in Sight) Registry 100542

Connor Ross, BS, Alexander Ivanov, MS, Tobias Elze, PhD, Joan W. Miller, MD, Flora Lum, MD, Alice C. Lorch, MD, MPH, Isdin Oke, MD, MPH, on behalf of the IRIS[®] Registry Analytic Center Consortium

There are geographic and temporal trends in missing data within the IRIS[®] Registry. Practice-level characteristics such as size, location, and patient population are associated with missing sociodemographic data and deserve consideration in registry-based analyses.



Circulating VEGF-A Levels in Relation to Retinopathy of Prematurity and Treatment Effects: A Systematic Review and Meta-Analysis 100548

Ulrika Sjöbom, MSc, Tove Hellqvist, BSc, Jhangir Humayun, MD, Anders K. Nilsson, PhD, Hanna Gyllenstein, PhD, Ann Hellström, MD, PhD, Chatarina Löfqvist, PhD

Systemic VEGF levels were reduced after treatment of retinopathy of prematurity. Serum shows a more pronounced reduction in VEGF levels than plasma after the intraocular injection of anti-VEGF.

Fluorescein Angiography Parameters in Premature Neonates 100561

Natasha F.S. da Cruz, MD, Sandra Hoyek, MD, Jesse D. Sengillo, MD, Ana Rodríguez, RN, RTT, Giselle de Oliveira, BFA, Catherin I. Negron, MBA, Nimesh A. Patel, MD, Audina M. Berrocal, MD

Fluorescein angiography timing in preterm infants can be variable and occur earlier than standard adult references.

(Continued)

Contents

(Continued)



Advancing Glaucoma Diagnosis: Employing Confidence-Calibrated Label Smoothing Loss for Model Calibration 100555

Midhula Vijayan, PhD, Deepthi Keshav Prasad, PhD, Venkatakrishnan Srinivasan, MTech

This study presents the Confidence-Calibrated Label Smoothing Loss method, enhancing machine learning model calibration in glaucoma diagnosis. This approach combines label smoothing with confidence penalties, improving model accuracy and reliability for clinical application.

Assessment of Retinal Volume in Individuals Without Ocular Disorders Based on Wide-Field Swept-Source OCT 100569

Yoshiaki Chiku, MD, Takao Hirano, MD, PhD, Ken Hoshiyama, MD, Yasuhiro Iesato, MD, PhD, Toshinori Murata, MD, PhD

In wide-field swept-source OCT images, age and left eye are negatively correlated with peripheral retinal volume. The thinnest part of the retinal quadrant differs between the macular and peripheral retinas.



Characterization of Retinal Microvascular Abnormalities in Birdshot Choroidopathy Using OCT Angiography 100559

Aman Kumar, MD, Alexander Zeleny, MD, Sunil Bellur, MD, Natasha Kesav, MD, Enny Oyeniran, MD, Kübra Gul Olke, MD, Susan Vitale, PhD, MHS, Wijek Kongwattananon, MD, H. Nida Sen, MD, MHS, Shilpa Kodati, MD

Quantifying blood flow in the retinal capillary plexus using OCT angiography in patients with birdshot choroidopathy reveals vessel density as an important biomarker in disease monitoring.

Genetic Reasons for Phenotypic Diversity in Neuronal Ceroid Lipofuscinoses and High-Resolution Imaging as a Marker of Retinal Disease 100560

Jennifer Huey, LCGC, Pankhuri Gupta, LCGC, Benjamin Wendel, MS, Teng Liu, MS, Palash Bharadwaj, PhD, Hillary Schwartz, BS, John P. Kelly, PhD, Irene Chang, MD, Jennifer R. Chao, MD, PhD, Ramkumar Sabesan, PhD, Aaron Nagiel, MD, PhD, Debarshi Mustafi, MD, PhD

Detailed genotypic and high-resolution phenotypic characterization provide insight into the varied retinal manifestations of neuronal ceroid lipofuscinoses.

(Continued)

Contents

(Continued)



Biological and Methodological Variability in Retinal Nerve Fiber Layer OCT: The Framingham Heart Study 100549

Louay Almidani, MD, MSc, Jasdeep Sabharwal, MD, PhD, Anoush Shahidzadeh, MPH, Ana Collazo Martinez, BS, Shu Jie Ting, MA, Brinda Vaidya, BS, Xuejuan Jiang, PhD, Tim Kowalczyk, MBA, Alexa Beiser, PhD, Lucia Sobrin, MD, MPH, Sudha Seshadri, MD, DM, Pradeep Ramulu, MD, PhD, Amir H. Kashani, MD, PhD

Retinal nerve fiber layer (RNFL) thickness is lower with older age, male gender, greater axial length, and lower signal strength (SS) while measurement variability is higher with greater average RNFL values and lower SS.



Quantitative Volumetric Analysis of Retinal Ischemia with an Oxygen Diffusion Model and OCT Angiography 100579

Pengxiao Zang, PhD, Tristan T. Hormel, PhD, Thomas S. Hwang, MD, Yali Jia, PhD

The authors proposed a 3-dimensional quantitative analysis method for oxygen tension and ischemia using an oxygen diffusion model with zero-order kinetics based on skeletonized vasculature in volumetric OCT angiography.

Sex Differences in Inflammation-Related Biomarkers Detected with OCT in Patients with Diabetic Macular Edema 100580

Xinyi Chen, MD, Wendy Yang, BS, Ashley Fong, MS, Noor Chahal, BS, Abu T. Taha, BS, Jeremy D. Keenan, MD, MPH, Jay M. Stewart, MD

Male sex was correlated with more inflammation-related biomarkers on OCT imaging in patients with diabetic macular edema, including more retinal hyperreflective foci, more hyperreflective choroidal foci, and a thicker inner nuclear layer.



Risk Factors for Focal Choroidal Excavation Concurrent with Chorioretinal Disease: Evaluated by Spectral-Domain OCT 100554

Yiwen Ou, MD, Minghui Qiu, MD, Mengyuan Li, MD, Yajun Mi, BS, Dezheng Wu, MD, PhD, Shibo Tang, MD, PhD, Weiwei Dai, MD, PhD, Jacey Hongjie Ma, MD, PhD

Focal choroidal excavation (FCE) is a rare clinical entity, in which the outer retina is indented into the choroid. The study is based on clinical demographic characteristics and imaging data to explore the risk factors between FCE and chorioretinal diseases. Moreover, the authors provide insight to elucidate the mechanism of the disease.

(Continued)

Contents

(Continued)

Interpretation of Clinical Retinal Images Using an Artificial Intelligence Chatbot 100556

Andrew Mihalache, MD(C), Ryan S. Huang, MD(C), David Mikhail, MD(C), Marko M. Popovic, MD, MPH, Reut Shor, MD, Austin Pereira, MD, Jason Kwok, MD, FRCSC, Peng Yan, MD, FRCSC, David T. Wong, MD, FRCSC, Peter J. Kertes, MD, CM, Radha P. Kohly, MD, PhD, Rajeev H. Muni, MD, MSc

A custom chatbot was able to accurately diagnose approximately half of the multimodal retina cases from a publicly accessible medical education platform, albeit relying heavily on text-based contextual information that accompanied ophthalmic images.

The Association between the Pulsatile Choroidal Volume Change and Ocular Rigidity 100576

Diane N. Sayah, OD, PhD, Denise Descovich, MD, Santiago Costantino, PhD, Mark R. Lesk, MD, MSc

The results of this study suggest that ocular rigidity could limit the amount of pulsatile blood entering the choroid, highlighting the interaction and dynamics between ocular blood flow and biomechanics in glaucoma.

Systemic Oxidative Stress Level as a Pathological and Prognostic Factor in Myopic Choroidal Neovascularization 100550

Jiying Wang, MD, PhD, Hiroshi Kunikata, MD, PhD, Masayuki Yasuda, MD, PhD, Noriko Himori, MD, PhD, Fumihiko Nitta, MD, PhD, Toru Nakazawa, MD, PhD

The oxidative stress level in eyes with high myopia (HM) differed according to the presence of myopic choroidal neovascularization (mCNV), subfoveal choroidal thickness, and requirements for intravitreal anti-VEGF antibody injection treatment. Measuring oxidative stress parameters might be useful in eyes with HM both for assessing mCNV risk and for predicting clinical outcomes in mCNV.



Censoring the Floor Effect in Long-Term Stargardt Disease Microperimetry Data Produces a Faster Rate of Decline 100581

Jason Chang, PhD, Jennifer A. Thompson, PhD, Rachael C. Heath Jeffery, MD, Amy Kalantary, MD, Tina M. Lamey, PhD, Terri L. McLaren, BSc, Fred K. Chen, PhD

Floor effect is a common confound in perimetry analysis. Progression rate is faster in Stargardt disease after accounting for floor effect, which can decrease duration or sample size in future clinical trials.

(Continued)

Contents

(Continued)



Comparison of Microperimetry and Static Perimetry for Evaluating Macular Function and Progression in Retinitis Pigmentosa 100582

Masatoshi Fukushima, MD, Yan Tao, MD, Sakurako Shimokawa, MD, Huanyu Zhao, MD, Shotaro Shimokawa, MD, PhD, Jun Funatsu, MD, PhD, Takahiro Hisai, MD, Ayako Okita, MD, PhD, Kohta Fujiwara, MD, PhD, Toshio Hisatomi, MD, PhD, Atsunobu Takeda, MD, PhD, Yasuhiro Ikeda, MD, PhD, Koh-Hei Sonoda, MD, PhD, Yusuke Murakami, MD, PhD

This study showed that both microperimetry and static perimetry are useful measures for assessing macular status; however, their usefulness decreased in assessing macular progression. This highlights the need for researchers to explore additional or more sensitive functional parameters to better monitor progression.



Reference Database for a Novel Binocular Visual Function Perimeter: A Randomized Clinical Trial 100583

Vincent Michael Patella, OD, Nevin W. El-Nimri, OD, PhD, John G. Flanagan, PhD, Mary K. Durbin, PhD, Timothy Bossie, OD, Derek Y. Ho, MD, PhD, Mayra Tafreshi, MBA, Michael A. Chaglasian, OD, David Kasanoff, OD, Satoshi Inoue, MSc, Sasan Moghimi, MD, Takashi Nishida, MD, PhD, Murray Fingeret, OD, Robert N. Weinreb, MD

A reference database was created for the TEMPO perimeter, emphasizing the importance of accounting for age and stimulus eccentricity when interpreting threshold visual field test results.



Prediction of Functional and Anatomic Progression in Lamellar Macular Holes 100529

Emanuele Crincoli, MD, FEBO, Barbara Parolini, MD, Fiammetta Catania, MD, FEBO, Alfonso Savastano, MD, Maria Cristina Savastano, MD, PhD, Clara Rizzo, MD, Raphael Kilian, MD, Veronika Matello, OD, Davide Allegrini, MD, Mario R. Romano, MD, PhD, Stanislao Rizzo, MD

Deep learning can accurately predict functional prognosis in lamellar macular holes. Ellipsoid zone damage, tissue loss, vessel density, and vessel length density in superficial capillary plexus and choriocapillaris flow deficit are the main imaging predictors of progression.



Special Commentary: My Perspective on Vision and Vision Rehabilitation 100532

August Colenbrander, MD

The author's career of >40 years involved clinical ophthalmology, visual science, and visual rehabilitation. Here, the author presents his personal views on how these domains overlap and interact and how each can contribute to successful rehabilitation.

(Continued)

Contents

(Continued)



Retinal Microvasculature Causally Affects the Brain Cortical Structure: A Mendelian Randomization Study 100465

Xiaoyue Wei, PhD, Wai Cheng Iao, MBBS, Yi Zhang, MBBS, Zijie Lin, MBBS, Haotian Lin, PhD

Retinal microvasculature causally affects the cortical structure, indicating that the retinal vascular parameters may serve as biomarkers for cortex pathologies and highlighting the potential of using noninvasive fundus images to predict neuropsychiatric disorders.



Is CONNECTDROP[®], a Medication Event Monitoring System Add-On Paired with a Smartphone Application, Acceptable to Patients with Glaucoma for Taking Their Daily Medication? The CONDORE Pilot Study 100541

Jean-Baptiste Dériot, MD, Emmanuelle Albertini, MD

A novel Medication Event Monitoring System connected to a Smartphone Application designed to improve treatment adherence in patients with glaucoma had high acceptability with patients and physicians in this proof-of-concept pilot study.

Case Reports

100558

Straylight of Explant Silicone Oil Samples to Predict Emulsification

Maximilian Hammer, MD, Leoni Britz, MD, Sonja Schickhardt, PhD, Donald Munro, BSc, Ramin Khoramnia, MD, Alexander Scheuerle, MD, Christian S. Mayer, MD, Philipp Uhl, PhD, Grzegorz Łabuz, PhD, Gerd Uwe Auffarth, MD, PhD



Ocular Manifestations of Enterovirus: An Important Emerging Pathogen

100562

Antony Boynes, MD, MMed, Chengde Pham, FRANZCO, Darren Jardine, PhD, Elsie Chan, FRANZCO

Correspondence (available at www.opthalmologyscience.org/current)

Re: Angela S Li et al.: Gradeability and Reproducibility of Geographic Atrophy Measurement in GATHER-1, a Phase II/III Randomized Interventional Trial

Siamak Sabour, MD, PhD, Fariba Ghassemi, MD



Reply

Angela S. Li, MD, Justin Myers, BS, Sandra S. Stinnett, DrPH, Dilraj S. Grewal, MD, Glenn J. Jaffe, MD

(Continued)

Contents

(Continued)

Myopia

A Novel Time-Aware Deep Learning Model Predicting Myopia in Children and Adolescents 100563

Ana Maria Varošaneć, MD, Leon Marković, MD, Zdenko Sonicki, MD, PhD

A 15-year retrospective study at the University Hospital “Sveti Duh” Zagreb analyzed 895 myopic children’s vision records to predict spherical equivalent using Extended Gate Time-Aware Long Short-Term Memory. Results showed a clinically acceptable error rate, aiding myopia intervention.

Posterior Eye Shape in Myopia 100575

Jost B. Jonas, MD, Songhomitra Panda-Jonas, MD, Zhe Pan, MD, Jie Xu, MD, Ya Xing Wang, MD

With longer axial length, the foveola is more often located outside of the geometrical posterior pole. It may be due to a meridional asymmetry in Bruch’s membrane enlargement in the fundus midperiphery.